

ASMEETA PRAKASH SAYAJI

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RESEARCH PROFILE

Master of Science in Computer Science with over 12 years of professional experience at the Space Applications Centre (ISRO), Ahmedabad. Through sustained independent study, I have developed a strong foundation in theoretical quantum computing, particularly quantum information theory, noisy quantum computation, quantum verification, and open quantum systems. I seek to pursue doctoral research combining theoretical analysis with computational modelling to understand the fundamental limits of quantum computation under realistic noise.

RESEARCH PROJECTS

Noise Modelling in Quantum Circuits

Investigated the influence of realistic noise on quantum computation by simulating amplitude damping, phase damping, depolarizing channels, and T_1/T_2 relaxation using Qiskit. Explored the impact of decoherence on quantum algorithm performance and simulation-based quantum error mitigation techniques, including Zero-Noise Extrapolation (ZNE).

Tools: Qiskit, Python, NumPy, Matplotlib

Open Quantum System Simulation

Investigated non-unitary quantum dynamics using Lindblad master-equation simulations in QuTiP to study decoherence and dissipative evolution in driven qubit systems.

Tools: QuTiP, Python

Quantum Algorithm Simulation

Implemented Bell-state preparation, Grover's Search, and Deutsch-Jozsa algorithms using Qiskit to investigate the behaviour of quantum circuits under ideal and noisy conditions.

Tools: Qiskit, OpenQASM

TECHNICAL SKILLS

Quantum Computing	Qiskit, QuTiP, OpenQASM, Quantum Circuit Simulation, Noise Modelling
Programming	Python, Java, C++, SQL, Git
Scientific Computing	NumPy, Matplotlib, LaTeX
Computer Science	Algorithms, Data Structures, Database Systems, Software Engineering

PROFESSIONAL EXPERIENCE

Senior Assistant - Procurement Division

Space Applications Centre (ISRO), Ahmedabad

Support the technical evaluation and procurement of specialised engineering systems used in scientific and space applications. Evaluate technical specifications and engineering compliance documents for cryogenic, microwave, RF, and scientific instrumentation. Review vendor technical documentation and assess compliance with institutional technical requirements. Coordinate technical evaluation activities with multidisciplinary engineering teams.

Assistant - Library & Information Division

Space Applications Centre (ISRO), Ahmedabad

Supported scientific information management, technical documentation, and access to scientific literature for researchers and engineers.

Private Tutor - Computer Science

Dhule, Maharashtra, India

Taught Bachelor of Science (B.Sc.) Computer Science students. Prepared instructional material and provided individual academic guidance. Developed strong communication and mentoring skills through teaching Computer Science concepts.

EDUCATION

Master of Science (M.Sc.) in Computer Science

July 2009

North Maharashtra University, India

Bachelor of Science (B.Sc.) in Computer Science, Physics & Mathematics

June 2006

North Maharashtra University, India

ACADEMIC ACHIEVEMENTS

Rank 1 - B.Sc. Computer Science

JNU Merit Scholarship

REFERENCES

Prof. Harshal S. Gavale

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